



Yarmouk University

Hijjawi Faculty for Engineering Technology

Computer Engineering Department

CPE 563: Computer Networking Laboratory

Task 6: VLANs

Basic Information:

Student Name:	
Student ID:	
Student Section Number:	
PC Logical Address:	
PC Physical Address:	

Objective:

The objective of this lab is to study how to divide a physical network into a number of separate logical networks using virtual local area networks (VLANs) with the benefit of decreasing collision domain and adding more security.

Background:

Virtual LANs (VLANs) allow a single extended LAN to be partitioned into several seemingly separate LANs. Each virtual LAN is assigned an identifier (sometimes called a color), and packets can only travel from one segment to another if both segments have the same identifier. This has the effect of limiting the number of segments in an extended LAN that will receive any given broadcast packet. An attractive feature of VLANs is that it is possible to change the logical topology without moving any wires or changing any addresses.

VLAN Configuration Guidelines

Follow these guidelines when creating and modifying VLANs in your network:

- The switch module supports 1005 VLANs.
- Normal-range Ethernet VLANs are identified with a number between 1 and 1001. VLAN numbers 1002 through 1005 are reserved for Token Ring and FDDI VLANs.

Creating or Modifying an Ethernet VLAN

To access VLAN configuration mode, enter the `vlan global configuration` command with a VLAN ID. Enter a new VLAN ID to create a VLAN, or enter an existing VLAN ID to modify that VLAN. You can use the default VLAN configuration (Table 11-2) or enter commands to configure the VLAN.

Beginning in privileged EXEC mode, follow these steps to create or modify an Ethernet VLAN:

Step	command
Step 1: Enter global configuration mode.	<code>configure terminal</code>
Step 2: Enter a VLAN ID, and enter VLAN configuration mode. Enter a new VLAN ID to create a VLAN, or enter an existing VLAN ID to modify that VLAN. The available VLAN ID range for this command is 1 to 4094.	<code>vlan vlan-id</code>
Step 3: (Optional and supported on normal-range VLANs only) Enter a name for the VLAN. If no name is entered for the VLAN, the default in the VLAN database is to append the <code>vlan-id</code> with leading zeros to the word VLAN. For example, VLAN0004 is a default VLAN name for VLAN 4.	<code>name vlan-name</code>
Step 4: Return to privileged EXEC mode	<code>end</code>
Step 5: Verify your entries. The name option is only valid for VLAN IDs 1 to 1005.	<code>show vlan {name vlan-name id vlan-id}</code>
Step 6: (Optional) Save the configuration in the switch module startup configuration file.	<code>copy running-config startup config</code>

To delete a VLAN, use the `no vlan vlan-id` global configuration command. You cannot delete VLAN 1 or VLANs 1002 to 1005.

Assigning Static-Access Ports to a VLAN

You can assign a static-access port to a VLAN.

Beginning in privileged EXEC mode, follow these steps to assign a port to a VLAN in the VLAN database:

Step	command
Step 1: Enter global configuration mode.	configure terminal
Step 2: Enter the interface to be added to the VLAN	interface interface-id
Step 3: Enable the port.	no shutdown
Step 4: Define the VLAN membership mode for the port (Layer 2 access port).	switchport mode access
Step 5: Assign the port to a VLAN. Valid VLAN IDs are 1 to 4094.	switchport access vlan vlan-id
Step 6: Return to privileged EXEC mode. switch module startup configuration file.	End
Step 7: Verify the VLAN membership mode of the	show running-config interface interface-id

Configuring VLAN Trunks

Trunking Overview

A trunk is a point-to-point link between one or more Ethernet switch module interfaces and another networking device such as a router or a switch module. Ethernet trunks carry the traffic of multiple VLANs over a single link, and you can extend the VLANs across an entire network. The switch module supports the 802.1Q industry-standard trunking encapsulation.

- If you do not intend to trunk across links, use the switchport mode access interface configuration command to disable trunking.
- To enable trunking, use the switchport mode trunk interface configuration command to change the interface to a trunk.

Configuring a Trunk Port

Beginning in privileged EXEC mode, follow these steps to configure a port as an 802.1Q trunk port:

Step	command
Step 1: Enter global configuration mode.	configure terminal
Step 2: Specify the port to be configured for trunking, and enter interface configuration mode.	interface interface-id
Step 3: Enable the port.	no shutdown
Step 4: Configure the interface as a Layer 2 trunk.	switchport mode trunk
Step 5: (Optional) Configure the list of VLANs allowed on the trunk. For explanations about using the add, all, except, and remove keywords, see the command reference for this release. The vlan-list parameter is either a single VLAN number from 1 to 4094 or a range of VLANs described by two VLAN numbers, the lower one first, separated by a hyphen. Do not enter any spaces between comma-separated VLAN parameters or in hyphen-specified ranges. All VLANs are allowed by default.	switchport trunk allowed vlan {add all except remove} vlan-list
Step 6: Return to privileged EXEC mode. switch module startup configuration file.	End
Step 7: Verify the VLAN membership mode of the	show running-config interface interface-id

Lab Work:

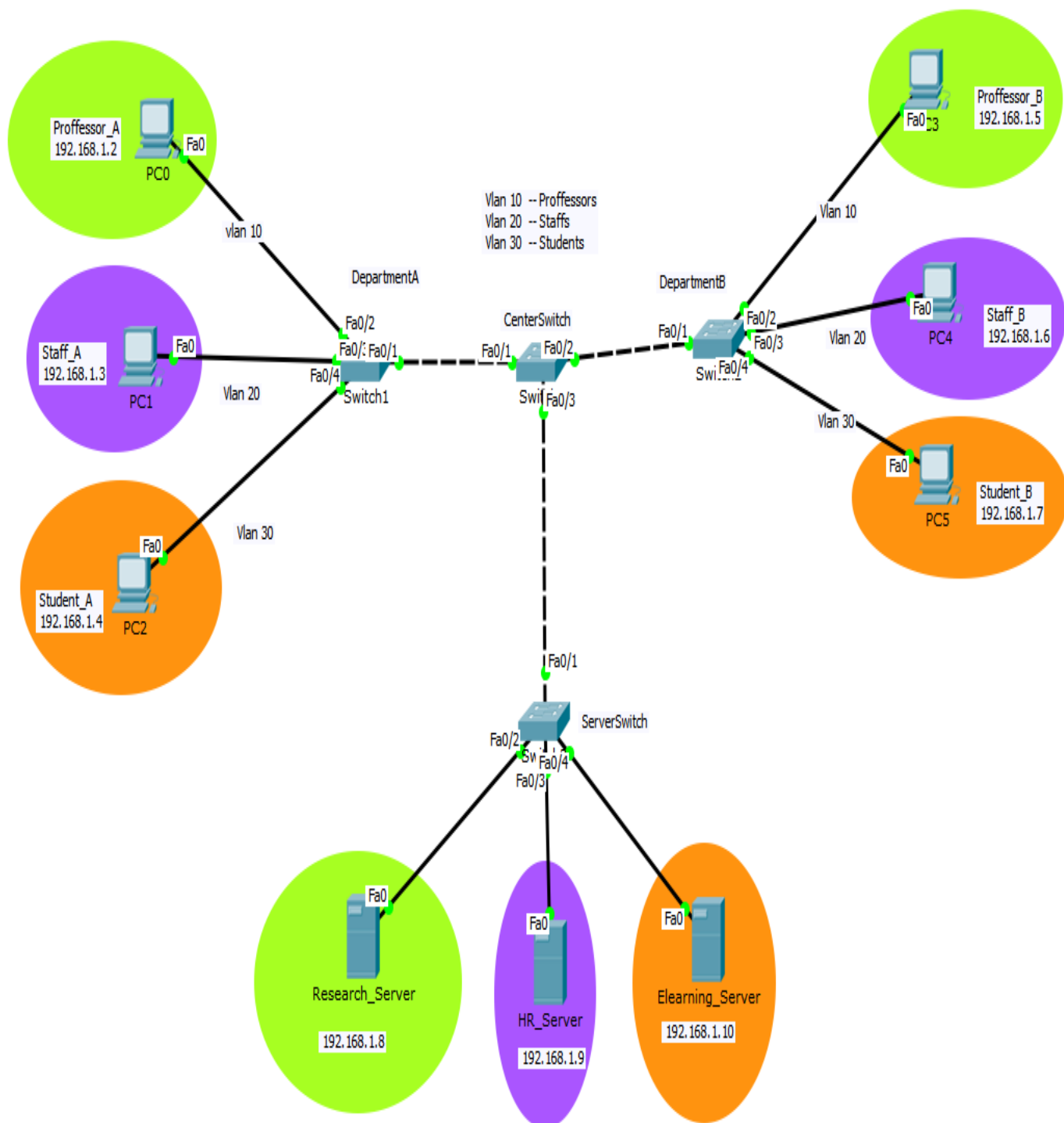


Fig 1

Referring to the above figure (Fig 1):

You have a network for a university with two departments A and B. Each department has a professor, a staff and a student. You should let the professors gain the access to the research server, and the staffs gain access to the HR_server and finally the students gain access to the ELearning server using VLANs.

While you are preparing your report, always provide a screenshot for each question.

Before doing VLANs configuration, answer the following questions:

1st. Try to ping from PC0 all other PCs, what you find? Why?

2nd. Try to ping from PC4 all the servers, does your trial succeed? Why?

After you finish VLANs configuration, answer the following questions:

3rd. For professors VLAN, test the connection between PC0, PC3 and the Research_Server, does it succeed?

4th. For Staffs VLAN, test the connection between PC1, PC4 and the HR_Server, does it succeed?

5th. For Students VLAN, test the connection between PC2, PC5 and the Elearning_Server, does it succeed?

6th. Using your own words, write your own observation about VLANs for this scenario.

Good Luck